Realized Volatility Forecasting with Time Series Foundation Models

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Abstract

Accurate forecasting of realized volatility is central to risk management, derivative pricing, and portfolio allocation. Recent work has applied a single time series foundation model to realized volatility, but no study compares multiple foundation models against standard econometric benchmarks using formal statistical testing. We evaluate two zero-shot time series foundation models—Chronos-Bolt and Moirai 2.0—against six econometric specifications (HAR, HAR-J, HAR-RS, HARQ, Log-HAR, ARFIMA) for four DJIA stocks across three forecast horizons (1, 5, and 22 days), using MSE, MAE, QLIKE, and out-of-sample R^2 as loss functions; statistical significance is assessed via Diebold–Mariano tests and Model Confidence Sets. Log-HAR achieves the lowest QLIKE at all horizons and is the only model in the Model Confidence Set for every asset at the daily horizon. Chronos-Bolt, however, delivers the highest R^2_{OOS} at every horizon, and the gap widens with the forecast period—at $h = 22$, its R^2_{OOS} exceeds Log-HAR's by 14.7 percentage points. Both foundation models enter the Model Confidence Set at weekly and monthly horizons, while all level-space HAR variants are excluded at the monthly horizon. Zero-shot foundation models add value for medium-

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and long-horizon volatility forecasting without domain-specific training, though log-space econometric models remain preferable under proportional loss criteria.

Keywords: Realized volatility, foundation models, time series forecasting, HAR, GARCH, machine learning

JEL Codes: C22, C53, C55, G17

1 Introduction

Volatility is a first-order input to option pricing, value-at-risk estimation, and portfolio optimization. Since the work of Andersen and Bollerslev (1998) and Barndorff-Nielsen and Shephard (2002), realized volatility constructed from high-frequency returns has served as the standard model-free measure of ex-post price variation. Forecasting realized volatility is now a central problem in financial econometrics, with direct implications for risk management, hedging, and capital allocation.

The Heterogeneous Autoregressive (HAR) model of Corsi (2009) established itself as the dominant benchmark. Its three-component structure—aggregating past realized volatility at daily, weekly, and monthly frequencies—provides a parsimonious approximation to the long-memory dynamics that characterize volatility. Subsequent extensions have incorporated jump variation (Andersen et al., 2007), semivariance asymmetries (Patton and Sheppard, 2015), and measurement error corrections (Bollerslev et al., 2016), yet the practical gains over the baseline specification have been mixed (Clements and Preve, 2021; Branco et al., 2024).

Machine learning methods—including random forests (Luong and Dokuchaev, 2018), neural networks (Bucci, 2020), and convolutional architectures (Moreno-Pino and Zohren, 2024)—have also been applied to realized volatility forecasting. Results are inconsistent: Bucci (2020) reports gains over HAR, while Branco et al. (2024) finds that linear models are difficult to outperform once the comparison accounts for multiple testing. A broader assessment by Christensen et al. (2023) concludes that machine learning offers incremental improvements but no paradigm shift for volatility prediction.

A different class of models has emerged over the past two years: time series foundation models (TSFMs), large pretrained transformers trained on billions of time series observations across domains. Leading examples include Chronos (Ansari et al., 2024), TimesFM (Das et al., 2024), and Moirai (Woo et al., 2024), each of which has since released second-generation versions with improved architectures and expanded training data (Ansari et al., 2025; Liu

et al., 2025a). These models can produce forecasts for previously unseen time series without any retraining—a capability known as zero-shot forecasting. Several surveys document the rapid pace of development in this area (Liang et al., 2024; Ye et al., 2024; Miller et al., 2024).

Applications of TSFMs to financial time series remain limited. Goel et al. (2025) tested TimesFM on realized volatility for 20 stocks and found that fine-tuning was necessary for the model to compete with HAR; zero-shot performance was poor. Rahimikia et al. (2025) evaluated Chronos, TimesFM, Moirai, and MOMENT on daily stock returns and reported uniformly negative zero-shot results, with simple ARIMA models dominating. However, realized volatility differs from returns in ways that may favor foundation models: it is strictly positive, exhibits mean reversion, displays long-memory dependence, and follows a distribution much closer to the macroeconomic and physical series on which TSFMs were pretrained. No study has yet evaluated multiple foundation models on realized volatility with formal statistical testing.

This paper makes five contributions. First, we provide the first multi-model TSFM comparison for realized volatility forecasting, evaluating Chronos-Bolt and Moirai 2.0 against six econometric benchmarks. Second, we test these models in a pure zero-shot setting—no domain-specific training—to isolate the value of general time series pretraining. Third, we analyze three forecast horizons ($h = 1, 5, 22$ days), uncovering horizon-dependent patterns in relative model performance. Fourth, we apply Model Confidence Set analysis (Hansen et al., 2011) to control for multiple comparisons, moving beyond pairwise tests. Fifth, we provide practical guidance on which model class to use under different loss functions and horizons.

Our results reveal a split verdict that depends on the loss function. Log-HAR achieves the lowest QLIKE at all three horizons and is the only model that survives the Model Confidence Set for every asset at $h = 1$ (Table 2). Under MSE and R^2_{OOS}, however, Chronos-Bolt dominates at every horizon, with the advantage growing as the forecast period lengthens—at $h = 22$, its R^2_{OOS} of 0.423 exceeds Log-HAR's 0.276 by 14.7 percentage points (Table 1). Both TSFMs enter the Model Confidence Set at $h \geq 5$, while all level-space HAR variants

are excluded at the monthly horizon. The choice of loss function therefore drives the model ranking: QLIKE, which penalizes proportional errors, favors log-space models; MSE, which penalizes absolute errors, favors the foundation models.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data and sample construction. Section 4 presents the econometric and foundation model specifications, along with the forecast evaluation framework. Section 5 reports the empirical results. Section 6 contains robustness checks, and Section 7 concludes.

2 Literature Review

2.1 Realized Volatility Modeling

The theory of realized volatility originates with Andersen and Bollerslev (1998) and Andersen et al. (2001), who showed that the sum of squared intraday returns provides a consistent, nonparametric estimator of the integrated variance of asset prices. Barndorff-Nielsen and Shephard (2002) formalized this result in a semimartingale framework, establishing that realized variance converges to integrated variance as the sampling frequency increases. Subsequent work identified the key stylized facts of realized volatility: approximate log-normality, long-memory dependence with a fractional integration parameter $d \approx 0.4$, and slow mean reversion (Andersen et al., 2003).

At ultra-high sampling frequencies, microstructure noise from bid-ask bounce and discrete price changes biases the realized variance estimator upward. Barndorff-Nielsen et al. (2008) introduced the realized kernel as a noise-consistent alternative that remains valid in the presence of market microstructure effects.

The HAR model of Corsi (2009) became the standard forecasting benchmark for realized volatility. By including daily, weekly, and monthly RV components as regressors, the model approximates long-memory behavior through a cascade of time scales corresponding

to heterogeneous trading horizons. The specification is parsimonious—three regressors plus a constant—yet achieves accuracy comparable to far more complex alternatives.

A large body of work has extended the HAR framework. Andersen et al. (2007) and Barndorff-Nielsen and Shephard (2004) separated realized variance into continuous and jump components using bipower variation, yielding the HAR-J model. Patton and Sheppard (2015) and Barndorff-Nielsen et al. (2010) decomposed realized variance into positive and negative semivariance to capture asymmetric responses to upside and downside moves. Bollerslev et al. (2016) introduced the HARQ model, which interacts the daily RV regressor with realized quarticity to account for time-varying measurement error. Clements and Preve (2021) provides a practical guide to implementing and comparing these extensions.

Long-memory models offer a complementary approach. ARFIMA models (Baillie et al., 1996; Izzeldin et al., 2019) explicitly parameterize the fractional integration order, while the rough volatility framework of Gatheral et al. (2018) models volatility as a fractional Brownian motion with Hurst exponent $H \approx 0.1$. Bennedsen et al. (2022) provides cross-asset evidence on the roughness of realized volatility, though the debate over whether roughness is genuine or an artifact of microstructure noise continues.

Machine learning methods have been applied to realized volatility forecasting with mixed results. Bucci (2020) reports that feedforward and recurrent neural networks outperform HAR on a sample of U.S. stocks. Luong and Dokuchaev (2018) finds gains from random forests. Moreno-Pino and Zohren (2024) introduces DeepVol, a convolutional neural network that processes raw order book data. In contrast, Branco et al. (2024) concludes that simple linear models are difficult to beat after correcting for multiple testing, and Christensen et al. (2023) finds that machine learning offers only incremental improvements over standard econometric methods. Recent surveys by Gunnarsson et al. (2024) and Leushuis and Petkov (2026) cover this literature.

2.2 Time Series Foundation Models

Time series foundation models are large pretrained models that produce forecasts for arbitrary time series without task-specific training, analogous to large language models for text (Liang et al., 2024). These models are trained on vast corpora of time series drawn from diverse domains—weather, energy, retail, transport, macroeconomics—and learn general temporal patterns that transfer to unseen series at inference time.

Three architectural families dominate the first generation of TSFMs. Chronos (Ansari et al., 2024) converts continuous values into discrete tokens via quantile binning and uses a T5 encoder-decoder architecture for autoregressive forecasting. TimesFM (Das et al., 2024) employs a patch-based decoder-only transformer with 200 million parameters, processing input in fixed-length patches and generating forecasts autoregressively. Moirai (Woo et al., 2024) handles arbitrary numbers of variates through an "any-variate" attention mechanism and uses mixture distribution outputs for uncertainty quantification.

Second-generation models appeared in 2025 with marked improvements. Ansari et al. (2025) introduced Chronos-2 and Chronos-Bolt: the former adds multivariate support and covariate handling, while the latter is a distilled encoder-only variant that produces direct quantile forecasts and runs up to 250 times faster than the original. Moirai 2.0 (Liu et al., 2025a) demonstrated that smaller, better-trained models can match or exceed their larger predecessors—a "less is more" finding—using multi-token prediction and improved tokenization. Other models include Lag-Llama (Rasul et al., 2024), MOMENT (Goswami et al., 2024), Timer (Liu et al., 2024), and Sundial (Liu et al., 2025b), each exploring different design choices for temporal representation and output distribution.

Standardized benchmarks have been developed to evaluate TSFMs across domains. GIFT-Eval (Aksu et al., 2024) provides a unified evaluation protocol across multiple datasets and forecast horizons. FEV-Bench (Shchur et al., 2025) focuses on forecasting evaluation with emphasis on calibration. TSFM-Bench (Li et al., 2024) compares models across zero-shot and fine-tuned settings. A consistent finding is that no single model dominates across all

domains and horizons, which motivates domain-specific evaluations such as the one we undertake here. Adler et al. (2025) raises additional concerns about the calibration of TSFM prediction intervals, finding systematic miscoverage across several models.

2.3 Foundation Models for Volatility Forecasting

The application of TSFMs to volatility forecasting is recent and limited. The closest prior work is Goel et al. (2025), who tested TimesFM 1.0 on realized volatility for 20 stocks. They found that zero-shot TimesFM underperformed HAR and that fine-tuning was necessary to achieve competitive accuracy. Our paper differs in several respects: we evaluate two foundation models from different families (Chronos-Bolt and Moirai 2.0), use newer model generations, and apply Model Confidence Set analysis to control for multiple comparisons. The same research group also applied TimesFM to Value-at-Risk forecasting (Goel et al., 2024).

Rahimikia et al. (2025) evaluated Chronos, TimesFM, Moirai, and MOMENT on daily stock returns and found that zero-shot TSFMs consistently underperformed ARIMA. However, daily returns differ from realized volatility in fundamental ways: returns are approximately symmetric, near-white-noise, and lack the strong serial dependence that characterizes realized volatility. Realized volatility is strictly positive, mean-reverting, and displays long-memory dynamics—properties that bring it closer to the macroeconomic and physical time series on which TSFMs were pretrained. This distinction suggests that negative results on returns need not carry over to realized measures.

Other financial applications include stock price forecasting (Łaniewski and Ślepaczuk, 2025; Valeyre and Aboura, 2024), foreign exchange volatility modeling (Nguyen et al., 2025), and multivariate financial time series (Marconi, 2025). Finance-specific foundation models have also been proposed, including FinCast (Zhu et al., 2025) and Kronos (Shi et al., 2025), though neither is publicly available at the time of writing, which justifies our focus on general-purpose models.

The question of whether to fine-tune or apply TSFMs zero-shot is itself an active area of research. Das et al. (2025) explored in-context fine-tuning for TimesFM, while Germán-Morales et al. (2024) and Gupta et al. (2024) investigated parameter-efficient adaptation via LoRA. Testing zero-shot performance first establishes an important baseline: if a TSFM already performs well without adaptation, the marginal value of fine-tuning may be limited, and the zero-shot workflow offers practical advantages in simplicity and computational cost.

We contribute to this emerging literature by providing the first evaluation that compares multiple time series foundation models against a broad set of econometric benchmarks for realized volatility, with formal statistical testing via Diebold–Mariano tests and Model Confidence Sets.

3 Data and Sample Construction

3.1 The VOLARE Dataset

[Describe the VOLARE dataset: Cipollini et al. (2026).]

3.2 Sample Selection and Descriptive Statistics

[Asset selection criteria, sample period, summary statistics table.]

4 Methodology

4.1 Econometric Benchmarks

We consider six econometric specifications that span the main approaches to realized volatility forecasting: the HAR family and its extensions, which exploit multi-scale aggregation, and ARFIMA, which explicitly models long memory.

HAR Model. The Heterogeneous Autoregressive model of Corsi (2009) captures the multi-scale persistence of realized volatility by aggregating past RV at daily, weekly, and monthly frequencies:

$$RV_{t+h} = \beta_0 + \beta_1 RV_t + \beta_2 RV_{t-5:t} + \beta_3 RV_{t-22:t} + \varepsilon_{t+h}, \tag{1}$$

where $RV_{t-k:t} = k^{-1} \sum_{i=0}^{k-1} RV_{t-i}$ denotes the average realized volatility over the previous k days. This three-component structure approximates long-memory behavior through the actions of heterogeneous market participants—short-term traders, medium-term portfolio managers, and long-term institutional investors—who operate at different time horizons (Corsi, 2009).

HAR-J Model. The HAR-J model augments the baseline HAR with a jump component to separate continuous and discontinuous variation (Andersen et al., 2007):

$$RV_{t+h} = \beta_0 + \beta_1 RV_t + \beta_2 RV_{t-5:t} + \beta_3 RV_{t-22:t} + \beta_J J_t + \varepsilon_{t+h}, \tag{2}$$

where $J_t = \max(RV_t - BPV_t, 0)$ measures the jump component as the positive part of the difference between realized variance and bipower variation (Barndorff-Nielsen and Shephard, 2004). A negative coefficient on J_t indicates that large jumps reduce, rather than increase, future volatility.

HAR-RS Model. The HAR-RS model decomposes realized variance into positive and negative semivariance components to capture asymmetric volatility responses (Patton and Sheppard, 2015):

$$RV_{t+h} = \beta_0 + \beta_1^+ RS_t^+ + \beta_1^- RS_t^- + \beta_2^+ RS_{t-5:t}^+ + \beta_2^- RS_{t-5:t}^- + \beta_3^+ RS_{t-22:t}^+ + \beta_3^- RS_{t-22:t}^- + \varepsilon_{t+h}, \tag{3}$$

where RS_t^+ and RS_t^- denote good and bad realized semivariance, respectively, and $RS_t^+ + RS_t^- = RV_t$ by construction (Barndorff-Nielsen et al., 2010). The six-regressor structure allows upside and downside risk to follow separate dynamics at each aggregation frequency.

HARQ Model. The HARQ model of Bollerslev et al. (2016) accounts for time-varying measurement error in realized volatility by interacting the daily RV regressor with the square root of realized quarticity:

$$RV_{t+h} = \beta_0 + (\beta_1 + \beta_1^Q \sqrt{RQ_t})\, RV_t + \beta_2 RV_{t-5:t} + \beta_3 RV_{t-22:t} + \varepsilon_{t+h}, \tag{4}$$

where RQ_t is the realized quarticity. The interaction term attenuates the daily RV signal when measurement noise is high, as indicated by large values of RQ_t.

Log-HAR Model. The Log-HAR model applies the HAR specification to log-transformed realized volatility (Andersen et al., 2003):

$$\log RV_{t+h} = \beta_0 + \beta_1 \log RV_t + \beta_2 \log RV_{t-5:t} + \beta_3 \log RV_{t-22:t} + \varepsilon_{t+h}. \tag{5}$$

The logarithmic transformation brings the RV distribution closer to normality, improving OLS efficiency. Point forecasts in levels are recovered via bias-corrected retransformation: $\widehat{RV}_{t+h} = \exp(\hat{y}_{t+h} + \hat{\sigma}^2/2)$, where $\hat{\sigma}^2$ is the estimated residual variance.

ARFIMA Model. The ARFIMA model captures the well-documented long-memory property of realized volatility through fractional differencing (Baillie et al., 1996):

$$(1-L)^d\, \Phi(L)\, (\log RV_t - \mu) = \Theta(L)\, \varepsilon_t, \tag{6}$$

where $d \in (0, 0.5)$ is the fractional integration parameter, $\Phi(L)$ and $\Theta(L)$ are autoregressive and moving average lag polynomials of orders p and q (with $p, q \leq 2$ selected by AIC), and

L is the lag operator. The parameter d is estimated by maximum likelihood. The fractional differencing filter is truncated at $K = 100$ lags.

All HAR-family models are estimated by OLS with Newey–West heteroskedasticity and autocorrelation consistent standard errors. The ARFIMA model is re-estimated every 22 trading days to balance computational cost against parameter staleness.

4.2 Time Series Foundation Models

We evaluate two time series foundation models (TSFMs) in a zero-shot setting—applied directly to realized volatility series without any domain-specific training or fine-tuning. Fine-tuning remains a direction for future work.[1]

Chronos-Bolt. Chronos-Bolt is a distilled variant of the Chronos family (Ansari et al., 2024, 2025). It uses an encoder-only transformer architecture and produces direct quantile forecasts without autoregressive decoding. We use the small checkpoint (`amazon/chronos-bolt-small`) and take the median (0.5 quantile) as the point forecast.

Moirai 2.0. Moirai 2.0 (Liu et al., 2025a) builds on the original Moirai architecture (Woo et al., 2024) with a decoder-only transformer and multi-token prediction. The small variant outputs nine quantile levels per forecast step. We use the `Salesforce/moirai-2.0-R-small` checkpoint and extract the median quantile as the point forecast.

Zero-shot protocol. Both TSFMs receive a rolling context window of 512 daily RV observations as input, with no additional features or covariates. For multi-step horizons ($h > 1$), the point forecast is the mean of the h-step output vector. No parameters are updated during the out-of-sample evaluation period; all forecasts are purely zero-shot.

[1] We originally planned to include TimesFM 2.5 (Das et al., 2024), but the publicly released model weights are incompatible with the current software package, preventing reliable inference.

4.3 Forecast Evaluation

Walk-forward design. We employ a walk-forward evaluation scheme with a rolling origin. The initial training window spans 252 trading days (approximately one year), after which we generate one-step-ahead forecasts over a test window of 126 days (approximately six months). The origin then advances by 126 days and the process repeats until the end of the sample. This produces approximately 4,834 out-of-sample forecasts per model–asset combination at $h = 1$. Foundation models follow the same walk-forward timing but require no estimation step; only the context window slides forward.

Forecast horizons. We evaluate three forecast horizons: $h = 1$ (one day), $h = 5$ (one week), and $h = 22$ (one month). The forecast target at horizon h is the average realized volatility over the next h days, $h^{-1} \sum_{i=1}^{h} RV_{t+i}$. All models produce direct h-step forecasts.

Loss functions. Our primary loss function is the quasi-likelihood (QLIKE) loss of Patton (2011):

$$L_{\text{QLIKE}}(\hat{\sigma}_t^2, RV_t) = \frac{RV_t}{\hat{\sigma}_t^2} - \log \frac{RV_t}{\hat{\sigma}_t^2} - 1, \tag{7}$$

where $\hat{\sigma}_t^2$ is the forecast and RV_t is the realized value. Patton (2011) shows that QLIKE preserves the ranking of forecasts regardless of the choice of volatility proxy, making it the preferred metric when the true conditional variance is unobserved. We also report mean squared error (MSE), mean absolute error (MAE), and out-of-sample R^2 (R^2_{OOS}). All forecasts are floored at 0.01 to prevent numerical instability in the QLIKE calculation.

Statistical tests. Pairwise forecast comparisons use the Diebold and Mariano (1995) test with Newey–West variance estimation, applied to QLIKE loss differentials. To address the multiple comparison problem inherent in evaluating eight models, we compute the Model Confidence Set (MCS) of Hansen et al. (2011). The MCS identifies the subset of models whose forecasting ability cannot be statistically distinguished from the best model at a

given significance level. We use the $T_{\max}$ test statistic with a block bootstrap (block length = 22 days, $B = 10{,}000$ replications) and significance level $\alpha = 0.10$.

5 Empirical Results

Table 1 reports average loss function values across the four sample assets (AAPL, AMZN, CAT, JPM) for each model and horizon. Table 2 presents Model Confidence Set inclusion rates.

Table 1: Aggregate forecast accuracy across four assets. Values are averages of MSE, MAE, QLIKE, and R^2_{OOS} across AAPL, AMZN, CAT, and JPM. Bold indicates the best value in each column within each panel. Lower is better for MSE, MAE, and QLIKE; higher is better for R^2_{OOS}.

Model	MSE	MAE	QLIKE	R^2_{OOS}
Panel A: $h = 1$ day				
HAR	31.57	1.45	0.299	0.480
HAR-J	31.64	1.45	0.542	0.480
HAR-RS	32.80	1.51	1.020	0.460
HARQ	35.76	1.47	13.189	0.423
Log-HAR	27.84	1.32	**0.133**	0.506
ARFIMA	17129.50	6.76	1.164	−530.5
Chronos-Bolt	**26.74**	1.27	0.147	**0.531**
Moirai 2.0	27.04	**1.25**	0.311	0.525
Panel B: $h = 5$ days				
HAR	19.33	1.52	0.429	0.507
HAR-J	19.47	1.52	0.449	0.504
HAR-RS	19.73	1.60	2.603	0.483
HARQ	24.51	1.59	15.349	0.334
Log-HAR	18.06	1.35	**0.152**	0.495
ARFIMA	17175.42	6.72	1.118	−840.5
Chronos-Bolt	**16.00**	1.25	0.170	**0.583**
Moirai 2.0	17.71	**1.24**	0.300	0.523
Panel C: $h = 22$ days				
HAR	19.93	1.90	4.149	0.250
HAR-J	20.28	1.90	4.369	0.244
HAR-RS	20.14	1.96	7.196	0.236
HARQ	1561.05	2.23	17.518	−26.1
Log-HAR	18.72	1.68	**0.239**	0.276
ARFIMA	17292.46	6.71	1.093	−1183.5
Chronos-Bolt	**16.14**	1.42	0.262	**0.423**
Moirai 2.0	17.27	**1.41**	0.351	0.352

Table 2: Model Confidence Set inclusion rates. Each cell shows the percentage of the four assets for which the model is included in the MCS at the 10% significance level, using QLIKE as the loss function. The MCS is computed with the $T_{\max}$ statistic, block bootstrap (block length = 22, B = 10,000).

Model	$h = 1$	$h = 5$	$h = 22$
HAR	0	75	0
HAR-J	0	50	0
HAR-RS	0	25	0
HARQ	0	0	0
Log-HAR	100	100	100
ARFIMA	0	0	0
Chronos-Bolt	0	75	75
Moirai 2.0	25	50	75

5.1 Aggregate Forecast Accuracy

The rankings differ across loss functions in a systematic way, reflecting the distinct penalty structures of MSE, MAE, and QLIKE (Table 1).

At the one-day horizon, Chronos-Bolt achieves the lowest MSE (26.74) and the highest R^2_{OOS} (0.531), outperforming Log-HAR (MSE = 27.84, $R^2_{\text{OOS}} = 0.506$) and all HAR variants. Moirai 2.0 produces the lowest MAE (1.25), narrowly beating Chronos-Bolt (1.27). Under QLIKE, however, Log-HAR dominates with a value of 0.133 compared to 0.147 for Chronos-Bolt and 0.311 for Moirai 2.0. The standard HAR model ranks fourth or fifth across most metrics, while its extensions (HAR-J, HAR-RS, HARQ) offer no consistent improvement.

The relative advantage of TSFMs grows at longer horizons. At $h = 5$, Chronos-Bolt's R^2_{OOS} of 0.583 exceeds Log-HAR's 0.495 by nearly nine percentage points, and its MSE (16.00) is 11% lower than Log-HAR's (18.06). At $h = 22$, this gap widens further: Chronos-Bolt achieves $R^2_{\text{OOS}} = 0.423$ versus 0.276 for Log-HAR, a difference of 14.7 percentage points. Meanwhile, the HAR-family models deteriorate sharply—HAR's QLIKE rises from 0.299 at $h = 1$ to 4.149 at $h = 22$, a degradation by more than an order of magnitude.

The divergence between MSE and QLIKE rankings is consistent across all three horizons.

QLIKE penalizes proportional forecast errors: underestimating a large RV observation incurs a heavier penalty than overestimating a small one by the same absolute amount. Log-HAR, by operating in log space, naturally produces forecasts that are proportionally well-calibrated, which explains its QLIKE advantage. The TSFMs, working in levels, minimize absolute deviations more effectively but produce occasional large proportional errors on extreme observations.

Two models exhibit instability. ARFIMA generates MSE values exceeding 17,000 at all horizons and deeply negative R^2_{OOS} (ranging from -530 to $-1{,}183$), indicating forecasts that are worse than a simple historical mean. This failure stems from poor out-of-sample behavior of the fractional integration parameter. HARQ also degrades at longer horizons, with MSE rising from 35.76 at $h = 1$ to 1,561 at $h = 22$, as the realized quarticity interaction term amplifies noise in multi-step forecasts.

5.2 Cross-Asset Heterogeneity

The aggregate rankings mask meaningful variation across individual assets. At $h = 1$, Chronos-Bolt achieves the lowest MSE for three of four assets (AAPL, CAT, JPM). CAT shows the largest TSFM advantage: Chronos-Bolt's MSE of 11.69 and R^2_{OOS} of 0.605 exceed HAR's 13.15 and 0.555 by wide margins. For AMZN, Log-HAR holds a QLIKE edge (0.114 vs. 0.122 for Chronos-Bolt), though the gap across all four metrics is narrow.

JPM illustrates the sensitivity of rankings to the choice of loss function. Moirai 2.0 achieves the lowest MAE for JPM at $h = 1$ (1.42 vs. 1.49 for Log-HAR), yet its QLIKE of 0.822 is more than five times Log-HAR's 0.146. This divergence arises because JPM exhibits occasional extreme volatility episodes where Moirai 2.0's level-space forecasts incur large proportional errors.

At $h = 22$, the TSFM advantage becomes more pronounced and extends to QLIKE for some assets. For CAT, Chronos-Bolt's QLIKE of 0.202 beats Log-HAR's 0.248, and Moirai 2.0 also outperforms Log-HAR (QLIKE = 0.207). For AMZN, Chronos-Bolt achieves

an R^2_{OOS} of 0.594 versus Log-HAR's 0.373, even though Log-HAR retains a slight QLIKE advantage (0.192 vs. 0.203). JPM remains the most difficult asset for TSFMs: Moirai 2.0's QLIKE of 0.681 is more than double Log-HAR's 0.294.

5.3 Statistical Significance and Model Confidence Sets

Table 2 reports the percentage of assets for which each model survives the MCS at the 10% significance level, using QLIKE as the loss function.

At $h = 1$, Log-HAR is the sole model that survives the MCS for all four assets. Moirai 2.0 enters the MCS for one asset (CAT), while all other models—including Chronos-Bolt—are excluded. Pairwise Diebold–Mariano tests confirm this pattern: for AAPL at $h = 1$, Log-HAR significantly outperforms both TSFMs ($p < 0.001$). The QLIKE advantage of Log-HAR at short horizons is statistically significant, not merely numerical.

At $h = 5$, the MCS expands. Log-HAR remains at 100%, but Chronos-Bolt and HAR both enter for 75% of assets, with HAR-J and Moirai 2.0 at 50%. The entry of Chronos-Bolt into the MCS at the weekly horizon indicates that the statistical gap between TSFMs and Log-HAR narrows as the forecast horizon increases.

At $h = 22$, the picture shifts further. Log-HAR again survives for all four assets, but both Chronos-Bolt and Moirai 2.0 now enter the MCS at 75%. Every HAR variant in levels—HAR, HAR-J, HAR-RS, HARQ—is excluded. For AAPL at $h = 22$, Log-HAR still outperforms Chronos-Bolt ($p = 0.021$), but the gap has narrowed substantially from $h = 1$ ($p < 0.001$). The monthly horizon represents the setting where zero-shot TSFMs add the most value: they match or approach the best econometric specification while requiring no parameter estimation.

6 Robustness Checks

[Alternative loss functions, sub-sample analysis, different volatility estimators, varying context lengths for TSFMs.]

7 Conclusion

[Summarize findings, discuss limitations, suggest directions for future research.]

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